POLITECNICO DI MILANO
High Performance Computing Engineering
Quantum Physics – July 15, 2025

Please justify each answer and use clear handwriting. Define each symbol you employ. For each exercise, give the symbolic solution and substitute numerical values (if required) only at the end. Write your first name, family name, and personal code on each sheet in a readable way. No electronic devices are allowed.

1. A quantum system is described by two energy eigenstates, $|1\rangle$ and $|2\rangle$, with corresponding energy eigenvalues E_1 and $E_2 > E_1$.
Now consider an operator $\hat{Q}$ that acts on these states as follows:

$$\hat{Q}|1\rangle = q(|1\rangle + i|2\rangle); \qquad \hat{Q}|2\rangle = q(|2\rangle - i|1\rangle),$$

with $q > 0$.

a) Show that $\hat{Q}$ is a Hermitian operator, find its eigenvalues and determine the corresponding eigenstates.
b) Find the expectation value of the observable associated to $\hat{Q}$ and its standard deviation when the system is in the ground state.

At $t = 0$ the system is in the eigenstate of $\hat{Q}$ corresponding to its higher eigenvalue.

c) Find the energy expectation value as a function of time.
d) Determine the time(s) at which the system can be found with certainty in the eigenstate of $\hat{Q}$ corresponding to its lower eigenvalue.

2. A particle with mass m is confined to move in one dimension along the x axis between the points $x = -a$ and $x = a$. The particle is subject to an infinite potential well such that its wavefunction vanishes outside this region.

a) Find all the possible results of a *single* measurement of the particle energy. Show that the energy eigenstates are either *even* or *odd* with respect to the origin of the x axis.

The particle wavefunction $\psi(x,t)$ at time $t = 0$ is $\psi(x, t = 0) = B(x^2 + C)$.

b) Determine the values of the parameters B and C and draw a qualitative plot for the probability density function associated with the particle position at time $t = 0$.
c) Find the probability that the particle is observed in the ground state or in the first excited state at time $t = 0$. Do these probabilities depend on time?
d) Determine the probability that the particle is found in the interval $-\frac{a}{2} \leq x \leq \frac{a}{2}$ at time $t = 0$. Discuss whether this probability changes over time.
[*Suggestion*: $\int_{-a}^{a}(x^2 - a^2)\cos\left(\frac{\pi}{2a}x\right)dx = -32\frac{a^3}{\pi^3}$]

Solution to the 1st exercise

a) In the representation based on the energy eigenstates, $\hat{Q}$ has the following matrix expression:

$$\hat{Q} = q\begin{bmatrix} 1 & i \\ -i & 1 \end{bmatrix}.$$

This matrix indeed corresponds to a Hermitian operator since $\hat{Q}^* = \hat{Q}_T$.
The normalized eigenstates of $\hat{Q}$ are:

$|q_1\rangle = \frac{1}{\sqrt{2}}(i|1\rangle + |2\rangle)$ with eigenvalue $2q$;
$|q_2\rangle = \frac{1}{\sqrt{2}}(-i|1\rangle + |2\rangle)$ with eigenvalue 0.

b) $\langle Q\rangle = q[1 \quad 0]\cdot\begin{bmatrix} 1 & i \\ -i & 1 \end{bmatrix}\cdot\begin{bmatrix} 1 \\ 0 \end{bmatrix} = q.$

$\sigma_Q = \sqrt{\langle Q^2\rangle - \langle Q\rangle^2}$, with $\langle Q^2\rangle = q^2[1 \quad 0]\cdot\begin{bmatrix} 1 & i \\ -i & 1 \end{bmatrix}\cdot\begin{bmatrix} 1 & i \\ -i & 1 \end{bmatrix}\cdot\begin{bmatrix} 1 \\ 0 \end{bmatrix} = 2q^2.$

From these expressions, one obtains $\sigma_Q = q$.

c) The state $|\psi(t)\rangle$, solution of the time-dependent Schrödinger equation satisfying the initial condition $|\psi(t=0)\rangle = |q_1\rangle$, is

$$|\psi(t)\rangle = \frac{1}{\sqrt{2}}\left(ie^{-i\frac{E_1 t}{\hbar}}|1\rangle + e^{-i\frac{E_2 t}{\hbar}}|2\rangle\right) = \frac{1}{\sqrt{2}}\begin{bmatrix} ie^{-i\frac{E_1 t}{\hbar}} \\ e^{-i\frac{E_2 t}{\hbar}} \end{bmatrix}.$$

From this expression, one can determine the energy expectation value:

$$\langle E\rangle = \frac{1}{2}\begin{bmatrix} -ie^{i\frac{E_1 t}{\hbar}} & e^{i\frac{E_2 t}{\hbar}} \end{bmatrix}\cdot\begin{bmatrix} E_1 & 0 \\ 0 & E_2 \end{bmatrix}\cdot\begin{bmatrix} ie^{-i\frac{E_1 t}{\hbar}} \\ e^{-i\frac{E_2 t}{\hbar}} \end{bmatrix} = \frac{E_1+E_2}{2},$$

which does not depend on time, as it should have been expected from energy conservation.

d) $\mathcal{P}[|q_2\rangle] = |\langle\psi(t)|q_2\rangle|^2 = 1 - \cos\left(\frac{E_2-E_1}{\hbar}t\right).$

This probability is equal to 1 when $\cos\left(\frac{E_2-E_1}{\hbar}t\right) = 0$, i.e., for $\frac{E_2-E_1}{\hbar}t = (2n+1)\frac{\pi}{2}$, with $n \in \mathbb{N}$. This condition implies $t = \frac{(2n+1)}{2(E_2-E_1)}\pi\hbar = \frac{(2n+1)}{4(E_2-E_1)}h$.

Solution to the 2nd exercise

a) The normalized energy eigenstates $\psi_n(x)$are:

$$\psi_n(x) = \begin{cases} \sqrt{\frac{1}{a}}\sin\left(\frac{n\pi}{2a}x\right) & n \text{ even } (n \neq 0); \\ \sqrt{\frac{1}{a}}\cos\left(\frac{n\pi}{2a}x\right) & n \text{ odd}, \end{cases}$$

which are indeed *even* (for $n =$ odd) or odd (for $n =$ even) because the Hamiltonian of the system commutes with the reflection operator $\hat{R}$ defined as $\hat{R}\varphi(x) = \varphi(-x)$.
The only observable values for the particle energy correspond to the following energy eigenvalues:

$$E_n = \frac{1}{2m}\left(\frac{n\pi\hbar}{2a}\right)^2.$$

b) C can be determined by noting that the wavefunction $\psi(x, t = 0)$ must be continuous at $x = \pm a$, which leads to $C = -a^2$.
B can be determined by the normalization condition

$$\int_{-a}^{a}|\psi(x, t = 0)|^2 dx = B^2 \int_{-a}^{a}(x^2 - a^2)^2 dx = B^2 \frac{16}{15}a^5 = 1,$$

which leads to $B = \sqrt{\frac{15}{16a^5}}$.

c) $\mathcal{P}[\psi_1(x)] = \left[\int_{-a}^{a}\psi(x, t = 0)\psi_1(x)dx\right]^2 = \left[\int_{-a}^{a}\psi(x, t = 0)\sqrt{\frac{1}{a}}\cos\left(\frac{\pi}{2a}x\right)dx\right]^2 = \frac{960}{\pi^6} \approx 0.999;$

$\mathcal{P}[\psi_2(x)] = \left[\int_{-a}^{a}\psi(x, t = 0)\psi_2(x)dx\right]^2 = \left[\int_{-a}^{a}\psi(x, t = 0)\sqrt{\frac{1}{a}}\sin\left(\frac{\pi}{a}x\right)dx\right]^2 = 0.$

The wavefunction $\psi(x, t = 0)$ is not an energy eigenfunction, meaning that $\psi(x, t)$ and, consequently, $\mathcal{P}[\psi_1(x)]$ vary with time.
Since the operator $\hat{R}$ commutes with the Hamiltonian, $\psi(x, t)$ remains *even* at any time. As a result, $\mathcal{P}[\psi_2(x)] = 0$ *at any time* because $\psi_2(x)$ is *odd*.

d) $\mathcal{P}\left[-\frac{a}{2} \leq x \leq \frac{a}{2}\right] = \int_{-a/2}^{a/2}|\psi(x, t = 0)|^2 dx = \frac{203}{256} \approx 0.8.$

For the reason discussed in the solution of the previous point, $\mathcal{P}\left[-\frac{a}{2} \leq x \leq \frac{a}{2}\right]$ will depend on time.